Title: Bayesian neural network with autoencoder for model-based description of α-particle preformation factor
Manuscript ID: CN10966

Response to the reviewer's comments and suggestions

We sincerely appreciate your constructive comments and suggestions, which are useful and have helped us to improve the quality of our work significantly. Please accept our renewed and heartfelt thanks for your thoughtful feedback.

General Assessment:

The authors have taken a positive step by providing a public code repository (github.com/Lin-lab-code/BNN); however, they have not included it in a dedicated Data Availability Statement within the manuscript. If the authors choose to make the code publicly available, several critical issues must be addressed. First, the project is currently not reproducible: the required dataset (BNN_input2.xlsx) is missing, and the provided scripts do not allow an independent reader to reproduce the figures and tables presented in the manuscript. Second, a close examination of BNN model.py reveals that the underlying computational implementation differs fundamentally from the manuscript's descriptions. This is not merely a matter of revising textual descriptions or updating initial parameter definitions; the code contains multiple hard-coded inductive biases and unfair baseline setups that raise serious concerns regarding the validity of the reported findings.

Response: We have uploaded the final version of the code and all input data to GitHub (github.com/Lin-lab-code/BNN), and have included a Data and Code Availability Statement in the manuscript to ensure that all figures and tables are fully reproducible. We have also made revisions and added clarifications to address points that may have been unclear or confusing in the previous version of the manuscript.

Main Issues to Fix

• **Missing Data and Lack of Reproducibility:** The script fails to run because the input dataset (BNN_input2.xlsx) is absent from the repository. Furthermore, the code outputs only a basic summary rather than generating the published results. The authors must provide the dataset alongside the complete scripts necessary to reproduce all figures and tables.

Response: We apologise for missing data and lack of reproducibility, and have addressed these issues carefully. We have uploaded the final version of the python source codes for the BNN network, and the direct executable Fortran file for α decay half-lives ($T^{\rm cal}$, $T^{\rm BNN-Auto}$ and $T^{\rm BNN-Auto_pre}$), as well as all the input data (from Phys. Rev. C 111, 034330 (2025), Phys. Rev. C 102, 044314 (2020), Chin. Phys. C 45, 030002 (2021), Chin. Phys. C 45, 030003 (2021), Chin. Phys. C 48, 074102 (2024)), to the GitHub repository (github.com/Lin-lab-code/BNN). We have double-checked that they can reproduce the results reported in the paper.

• **Contradictory Network Architecture:** The manuscript states that the prediction module uses a 24-dimensional input, whereas the implementation in the code specifies a 16-dimensional input. Additionally, while the authors stated in their response that the decoder output is excluded from the prediction module, the code explicitly feeds this reconstructed output (recon) into the shared prediction layers, indicating a fundamentally different architecture.

Response: Sorry for the confusion. Our manuscript adopts a 24-dimensional fusion input structure. In the code, for the standardized 8-dimensional input x, the encoder first generates an 8-dimensional latent representation z. In parallel, the original input passes through an independent skip-mapping branch to produce another set of 8-dimensional representations s(x). The latent representation, the skip-mapping representation, and the standardized original input are then concatenated to form a 24-dimensional fusion representation, denoted as $h_0 = [z, s(x), x] \in R^{24}$. This 24-dimensional fused representation is fed into the Bayesian prediction module, whose architecture is $24 \to 128 \to 64 \to 1$. The reconstructed output (recon) of the decoder does not enter the prediction module; rather, it is constrained solely by the reconstruction loss to preserve useful information in the latent representation derived from the input physical features. We have explained this issue more clearly in the updated manuscript.

• **Hidden Parity Grouping:** The manuscript concludes that the model naturally captures the odd-even staggering effect. However, the code manually partitions nuclei by parity (even-even, odd-A, odd-odd) and routes them into three hard-coded, separate "expert" subnetworks. This manual separation introduces a substantial structural shortcut, which invalidates the claim that the model learned this physical phenomenon autonomously.

Response: We apologize for having uploaded an earlier version rather than the final version, which led to this confusion. We have now confirmed that the correct version has been uploaded to GitHub, which can naturally capture the odd-even staggering effect.

• **Unfair Baseline Comparisons:** The proposed BNN model is provided with explicit parity grouping during training, whereas the baseline models (Random Forest, SFNN, etc.) are restricted to the raw features alone. For the comparison to be scientifically sound and fair, all models must be trained and evaluated using identical input features and structural information.

Response: We fully agree that, for a comparison to be scientifically sound, all models must be trained and evaluated using the same input raw data and features. We have stated this issue explicitly in the updated manuscript.

• **Inverted Parameter-to-Sample Ratio:** The authors report a parameter-to-sample ratio of 1:91.5 to argue that the model resists overfitting. This reporting appears inverted and misleading: with approximately 44,000 parameters and only 481 samples, there are actually roughly 92 parameters per individual sample, which points to extreme overparameterization rather than the converse.

Response: Thank you very much for pointing out this typo. We have corrected it in the updated manuscript.

We hope that the revised manuscript and responses, together with the reproducible code repository, will fully address your concerns. Thank you again for your valuable time and insightful suggestions.

Sincerely yours,

Xiao-Yan Zhu on behalf of all the authors.